

Chameleon

Development changelog — major changes across all build checkpoints

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This document summarizes every major change made to the Chameleon Chrome extension across its full build history: the original scaffold, the Chameleon rebrand, and all six development checkpoints. Each checkpoint was verified live in a real browser (Edge + Playwright) before being marked complete, not just written and assumed to work.

Phase 0 — Original build

"Accessibility Intelligence for Product Teams"

- Manifest V3 extension: activeTab/scripting/storage permissions only, no broad host permissions.
- axe-core bundled locally, injected on demand via chrome.scripting — never loaded on every page.
- Shadow DOM floating widget + panel, fully isolated from host page styles in both directions.
- Issues tab: axe + custom checks, weighted severity x reach x occurrence x critical-path scoring, POUR/component grouping, click-to-highlight.
- Fixes tab: deterministic (non-LLM) color-contrast math, target-size/label/name/alt-text generators, live preview + undo, Copy patch.
- Executive tab: trend sparkline, top risks, legal-exposure note, downloadable HTML report.
- Roadmap tab (display-only), demo.html broken-checkout test page, README.

Two real bugs found during live verification and fixed in the shipped source: widget.css missing from web_accessible_resources (widget rendered unstyled), and a double ":1:1" bug in the contrast-ratio message text.

Phase 0.5 — Rebrand + demo-readiness pass

- Renamed Accessibility Intelligence -> Chameleon across manifest, panel UI, README.
- New severity/score formula: Critical=10 / Serious=5 / Moderate=2 / Moderate(minor)=1 weights, penalty = weight x min(instances,3), 0-100 clamped score, 4 verdict bands, before/after delta.
- Color-blind-safe severity palette (vermilion/amber/blue) replacing the original red/orange/gold scheme.
- Task blockers summary + "Fix first" priority stepper added to the Issues tab.
- Severity-colored, non-flashing pulse highlight for "Highlight on page," replacing the fixed-orange outline.
- Demo safety: a failed re-scan now falls back to the last good scan, labeled "Cached result from [time]" instead of a bare error.
- DEMO.md added: timed 2-minute script + live recovery plan.

Checkpoint 1 — Page style manager + Dyslexia mode

- New style-manager.js: the single module spec-required for every page-altering feature, owning one <style id="chameleon-page-style"> element with 6 defined precedence levels, idempotent rebuild, and withSuspended() so scores always describe the real page.
- Dyslexia-friendly font mode wired as the first real consumer: header toggle, persisted via chrome.storage.local.
- "Reset page" button added to the header.

Verified: Live run confirmed: toggling Dyslexia changed body font-family with exactly one managed <style> tag; the panel itself stayed completely unaffected (screenshot showed page in Comic Sans, panel in its own font) because the host lives outside <body>; Reset page fully reverted everything.

Checkpoint 2 — Color Lab tab

- New Color Lab tab: text/background color pickers + synced hex inputs, sliders (font size, line height, letter/word spacing), live preview, live WCAG contrast ratio with AA/AAA pass-fail (normal + large text).
- "Suggest accessible colors" reusing the same deterministic contrast math as the Fixes tab.
- Color-blind simulation (Protanopia/Deuteranopia/Tritanopia/Achromatopsia) via SVG filter, applied page-wide as an explicitly-labeled preview, never a fix.
- Low-vision tools: Magnify (100-400% with horizontal-overflow detection), cursor-size boost, high-contrast mode (B/W, yellow/black, invert), focus booster.

Verified: Live-measured: default contrast 16.58:1 (all pass) -> changed to #aaaaaa updated live to 2.32:1 (all fail); Apply to page set

real computed body color; color-blind filter confirmed applied to the page while the panel itself stayed fully colorful (Shadow DOM exclusion held under a page-wide CSS filter, not just plain styles); Magnify 200% -> computed zoom:2; full Reset page reverted color/filter/zoom together.

Checkpoint 3 — Confidence labels + Needs-manual-review

- Confidence labels: every issue now Confirmed or Possible. Four new heuristic checks (vague link text, sound-only language, reading level, key-flow detection) run as Possible findings — softer wording, capped severity, never scored, Confirm/Dismiss actions.
- Needs-manual-review merges axe's own "incomplete" results with the static review list, adds a count badge and per-item "Mark reviewed."
- Issue states: Open/Ignored/Fixed/Regressed, persisted per rule per URL in chrome.storage.local. Auto-detects "Fixed since last scan" by diffing issue IDs across scans; a re-appearing "fixed" issue flips back to Open with a Regressed badge.
- Filter chips: All / Critical / Serious / Moderate / Needs review / Ignored / Fixed, each with a live count.

Verified: A real bug was caught and fixed mid-checkpoint: the first implementation re-ran a full axe scan after "Mark fixed," which immediately re-detected the still-present issue and falsely flipped it to Regressed. Fixed by making Ignore/Fixed/Confirm/Dismiss local, instant state mutations instead of real re-scans. Verified live: chip counts summed correctly, Mark fixed moved an issue instantly with zero false regressions, Fixed-filter view rendered the correct entry with a working Reopen button.

Checkpoint 4 — Read Aloud + Sound alternatives + Screen

- New "What a screen reader hears" tab: independent reading-order walker listing role + accessible name for every heading/landmark/link/button/field, with inline problem flags, explicitly labeled an approximation.
- Sound alternatives header toggle: badges every <video>/<audio> with caption/transcript status (extended style-manager.js with a non-CSS apply/remove pair, still owned by that one module); Chameleon's own announcements also surface as a visible on-page toast when this mode is on.
- Read Aloud header toggle using speechSynthesis, restricted to localService voices only (stays off with a plain message if none exist, so nothing is ever sent to a remote voice). Play/Pause/Resume/Stop, speed, voice selection, per-card "Read this issue," on-demand page-content reading, current-item highlighting.

Verified: Live run found 3 real local Windows voices (Microsoft David/Mark/Zira); confirmed speechSynthesis.speaking:true after toggling on; confirmed the visible toast updates correctly even on an async-triggered announce (storage round-trip -> announce), not just the synchronous case; Screen Reader tab detected 23 landmarks/controls with 8 correctly flagged as missing a name.

Checkpoint 5 — Exports (PDF / JSON / CSV / SARIF) + Co

- PDF report: vendored jsPDF locally (same local-injection pattern as axe-core), multi-page report covering scores+formula, severity counts, every confirmed issue with its fix, Possible findings, Needs-manual-review, Ignored/Fixed, a screen-reader summary, and the Roadmap. Filename follows chameleon-report-YYYY-MM-DD-pagename.pdf.
- JSON export with a stable schema (rule id, severity, selector, WCAG criterion, confidence, state, fix words/diff).
- SARIF 2.1.0 export (Critical/Serious -> error, Moderate -> warning) for GitHub code scanning.
- CSV export, Copy-summary-to-clipboard, and a per-card "Copy for developer" ticket button.
- FUTURE.md added: planned CLI + GitHub Action note, explicitly labeled not-built.

Verified: Downloaded files were actually read back and validated, not just confirmed to exist: JSON had 11 confirmed issues at schemaVersion 1.0; CSV had 12 lines with the correct header; SARIF reported version 2.1.0 with 11 results; the downloaded PDF's bytes genuinely started with %PDF- at 18KB. Clipboard content for both Copy actions was read back and matched expectations.

Checkpoint 6 — Tab bar, launcher pulse, Position/themes

- Fixed the clipped tab bar (flagged in Checkpoint 4): switched from forced-fit flex:1 tabs to natural sizing + horizontal scroll.
- Launcher pulse badge: soft ring pulse (logo color never changes) on new Critical issues, max 3 pulses, prefers-reduced-motion safe, settles to a static count badge.
- Position menu: dock Left/Right/Bottom/Floating, persisted per site; arrow-key nudge (Shift = bigger step) as the keyboard-operable alternative to dragging. Real mouse drag/resize/minimize explicitly not built.
- Keyboard tab-order audit overlay (numbered badges in real tab order).
- Light/dark/high-contrast panel chrome themes via CSS custom properties, persisted; logo keeps a readable backing in all three.
- First-run 3-step tour; generic POUR-keyed "Learn more" example per issue card.

- Self-check dev script (scripts/self-check.js) running axe-core against the panel itself.
- ARCHITECTURE.md, PITCH.md, ACCESSIBILITY-TESTING.md added.

Verified: Launcher badge showed the exact live Critical count (4) with a matching aria-label. Docking Right set class=dock-right; two ArrowLeft presses while floating moved the panel from left:844.8 to 804.8 -- exactly -40px (2 x 20px step). Dark theme set data-theme=dark with computed panel background rgb(27,31,35), the exact defined value. The self-check script was actually executed against the real build: 0 total violations, 0 Critical/Serious, on Chameleon's own panel.

Logo integration

- Real logo art (cham-logo-demo.png, 2000x2000) processed into assets/icons/ at 16/32/48/64/96/128px via high-quality bicubic downscaling.
- Replaced the placeholder inline SVG in the launcher and panel-header title with the real logo image, served from the extension bundle (chrome.runtime.getURL), declared in web_accessible_resources.
- manifest.json icons/action.default_icon updated to the new assets/icons/ paths; old placeholder icon generator and output removed.

Known limitations (explicitly deferred, not forgotten)

- Real mouse-drag-to-reposition, drag-to-resize, and minimize/collapse for the panel — keyboard-operable Position menu + arrow-key nudge were built instead.
- Full rules-catalog.json / user-groups.json consolidation — lookup.js already functionally serves this role; merging it with scoring.js/fixes.js into one new file was judged a high-risk, low-user-value refactor this late in the build.
- Per-rule "Learn more" examples — shipped 4 generic POUR-level examples instead of ~70 rule-specific ones.
- Visible-focus-indicator automated detection — would require programmatically focusing every interactive element, risking side effects on the host page; left as a manual checklist item in ACCESSIBILITY-TESTING.md instead.
- Live on-device captions — would require a remote speech-recognition API, breaking the "nothing leaves your browser" guarantee.
- CLI / GitHub Action for CI — planned and sketched in FUTURE.md, not built.